1.7.11: Combining symmetry

Only certain combinations of symmetry operation can exist in a crystal structure. This is because one symmetry element operating on another will generate a third symmetry element in the structure and this can end up generating an infinite number of symmetry elements, as shown in the animation below:

In fact, there are only 32 permitted combinations of mirror planes, rotation axes, centres of symmetry and inversion axes. These are known as the 32 **point groups**. Each point group is a finite set of mutually compatible symmetry elements. When the symmetry elements of a point group are operated on each other, they simply generate one of the other elements within the group.